

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangam, Machhe, Belagavi-590018



Scheme of Teaching and Examinations 2025

Outcome-Based Education (OBE) and Choice-Based Credit System (CBCS)
(Effective from the academic year 2025-26)

B.E. in Medical Electronics Engineering
Scheme of Teaching and Examinations 2025

Outcome-Based Education (OBE) and Choice-Based Credit System (CBCS) (Effective from the academic year 2025-26)

III SEMESTER															
Sl. No	Course	Course Code	Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)		Teaching & Learning Scheme				Examination				Credits	
						CI		LI	TW & SL	#	Duration in hours	CIE Marks	SEE Marks		Total Marks
						L	T	P	SL						
1	ASC/PCC	1BMATEC301	Transform Techniques and Optimization Theory	Maths Dept		42	28	0	50	120	3	50	50	100	4
2	IPCC	1BEC302	Digital System Design Using Verilog	Respective Engg Dept		42	0	28	50	120	3	50	50	100	4
3	PCC	1BEC303	Network Analysis	Respective Engg Dept		42	28	0	50	120	3	50	50	100	4
4	PCC	1BEC304	Analog Electronics and Linear Integrated Circuits	Respective Engg Dept		42	0	0	48	90	3	50	50	100	3
5	PCC	1BML305	Human Anatomy and Human Physiology	Respective Engg Dept		42	0	0	48	90	3	50	50	100	3
6	PCCL	1BECL306	Analog Electronics and Linear Integrated Circuits Lab	Respective Engg Dept		0	0	28	2	30	2	50	50	100	1
7	AEC	1BMLL307x	Ability Enhancement Course Laboratory**	Respective Engg Dept		0	0	28	2	30	2	50	50	100	1
8	SDC	1BCP308	Community Project (Project-Based Learning) / Societal Project	Respective Engg Dept		0	0	0	30	30	2	50	50	100	1
9	NCMC	1BNSS309	National Service Scheme (NSS)	Campus	NSS coordinator	0	0	0	28	28	--	100	---	100	PP
		1BPE309	Physical Education (PE) (Sports and Athletics)		Physical Education Director										
		1BYOG309	Yoga		Yoga Teacher										
		1BMUK309	Music		Music Teacher										

Total											500	400	900	21
10	NCMC	1BMATDIP310	Mathematics course for Lateral Entry Students	Maths Dept	14	0	0	14	28	3	100	---	100	PP
Total									686		600	400	1000	21

TH/S- Total Hours per Semester, TW & SL- Term Work & Self Learning

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Ability Enhancement Course (Laboratory) 1BMLL307x			
1BMLL307A	Analog Electronic Circuit Simulations Lab	1BMLL307C	Introduction to Data Structures and OOPs using Python Lab
1BMLL307B	Python Programming Lab	1BML307D	Programming using JAVA

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IV SEMESTER															
Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)		Teaching & Learning Scheme					Examination				Credits
						CI		LI	# TW & SL	T	D	CIE Marks	SEE Marks	Total Marks	
						L	T	P							
1	ASC/PCC	1BMATEC401	Mathematics for Machine Learning	Maths Dept	42	0	0	48	90	3	50	50	100	3	
2	IPCC	1BEC402	Applied Computer Organization and Microcontrollers	Respective Engg Dept	42	0	28	50	120	3	50	50	100	4	
3	PCC	1BML403	Basics of Signals and Systems	Respective Engg Dept	42	28	0	50	120	3	50	50	100	4	
4	PCC	1BML404	Medical Physics	Respective Engg Dept	42	0	0	48	90	3	50	50	100	3	
5	PCCL	1BMLL405	Medical Physics Lab	Respective Engg Dept	0	0	28	02	30	2	50	50	100	1	
6	AEC	1BMLL406	Ability Enhancement Course Laboratory**	Respective Engg Dept	0	0	28	02	30	2	50	50	100	1	
7	BSC	1BEC407	Biology for Electrical and Electronics Engineers	Phy/Che/Respective Engg Dept.	28	0	0	32	60	3	50	50	100	2	
8	SDC	1BEP408	Environmental Science Project	Respective Engg Dept	0	0	0	30	30	3	50	50	100	1	
9	PCC/PEC	1BML409	Biomedical Sensors and Measurements	Respective Engg Dept	42	0	0	48	90	3	50	50	100	3	
10	NCMC	1BNSK409	National Service Scheme (NSS)	C a m p u s	0	0	0	28	28	--	100	---	100	PP	
		1BPEK409	Physical Education (PE) (Sports and Athletics)												
		1BYOK409	Yoga												
		1BMUS409	Music												
Total									598		550	450	1000	22	
11	NCMC	1BMATDIP41	Mathematics course for Lateral Entry	TD/PSB -Maths Dept	14	0	0	14	28	--	100	--	100	PP	

		0	Students											
Total									626		650	450	1100	22
	Ability Enhancement Course (Laboratory) 1BMLL406x													
1BMLL406A	IOT lab				1BMLL406C	Signals Lab								
1BMLL406B	Biomedical Sensors and Measurements Lab				1BMLL406D	Virtual Bioinstrumentation Lab								

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B.E. in Medical Electronics Engineering Scheme of Teaching and Examinations 2025 Outcome-Based Education (OBE) and Choice-Based Credit System (CBCS) (Effective from the academic year 2025-26)														
V SEMESTER														
Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)	Teaching & Learning Scheme				Examination				Credits	
					CI		LI	# TW & SL	# TH/S	D u r a t i o n i n h o u r s	CIE Marks	SEE Marks		Total Marks
					L	T	P							
1	HSMC	1BML501	This course must be pertaining to economics and management of the concerned degree program. The course syllabus should have both economics and management topics and the course title should bear the word Management.	Respective Engg Dept	42	0	0	48	90	3	50	50	100	3
2	IPCC	1BML502	Biomedical Signal Processing	Respective Engg Dept	42	0	28	50	120	3	50	50	100	4
3	PCC	1BML503	Control Systems	Respective Engg Dept	42	28	0	20	90	3	50	50	100	3
4	PCC	1BML504	Diagnostic and Therapeutic Equipment	Respective Engg Dept	42	0	0	48	90	3	50	50	100	3
5	PEC	1BML505x	Professional Elective Course-I	Respective Engg Dept	42	0	0	48	90	3	50	50	100	3
6	BSC	1BRM506	Research Methodology and IPR	Respective Engg Dept	28	0	0	32	60	02	50	50	100	2
7	PCCL	1BMLL507	Diagnostics and Therapeutic Equipment Lab	Respective Engg Dept	0	0	28	2	30	02	50	50	100	1
8	SDC	1BML508	Hackathon-Based Project	CIE: By Departments	0	0	0	60	60	--	50	50	100	2

				SEE: Evaluation by industry experts											
	Total									630		400	400	800	21
	Professional Elective Course-I														
1BML505A	Introduction to Medical Electronics			1BML505C	Data Communication in Healthcare										
1BML505B	Medical Informatics			1BML505D	Augmented/ Virtual Reality in Healthcare										

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	VI SEMESTER													
Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)	Teaching & Learning Scheme					Examination				Credits
					CI		LI	# TW & SL	# TH / S	Duration in hours	CIE Marks	SEE Marks	Total Marks	
					L	T	P							
1	IPCC	1BML601	VLSI Design	Respective Engg Dept	42	0	28	50	120	3	50	50	100	4
2	PCC	1BML602	Biomedical Image Processing	Respective Engg Dept	42	28	0	20	90	3	50	50	100	3
3	PCC	1BML603	Medical Device Design & Development	Respective Engg Dept	42	0	0	48	90	3	50	50	100	3
4	PCC	1BML604	Machine Learning for Medical Engineering	Respective Engg Dept	42	0	0	48	90	3	50	50	100	3
5	PEC	1BML605x	Professional Elective Courses-II	Respective Engg	42	0	0	48	90	3	50	50	100	3

				Dept											
6	PCCL	1BMLL606	Biomedical Instrumentation Lab	Respective Engg Dept	0	0	28	2	30	2	50	50	100	1	
7	AEC	1BMLL607x	Ability Enhancement Course Laboratory	Respective Engg Dept	0	0	28	2	30	2	50	50	100	1	
8	SDC	1BML608	Capstone Project - Phase I	Respective Engg Dept	0	0	0	50	90	3	100	--	100	3	
9	NCMC	1BML609	Universal Human Value	TD/PSB	14	0	0	14	28	--	100	---	100	PP	
	Total								630		550	350	900	21	
	Professional Elective Course-II														
1BML605A		Medical Equipment and Trouble Shooting			1BML605C		Artificial Neural Networks & Deep Learning								
1BML605B		Biostatistics			1BML605D		Data Mining and Its Applications in Medicine								
	Ability Enhancement Course Laboratory**														
1BMLL607 A		Basics of Machine Learning Lab			1BMLL607C		Biomedical Image Processing Lab								
1BMLL607 B		Biostatistics Lab			1BMLL607D		Tools for Machine Learning and Deep Learning								
B.E. in Medical Electronics Engineering Scheme of Teaching and Examinations 2025 Outcome-Based Education (OBE) and Choice-Based Credit System (CBCS) (Effective from the academic year 2025-26)															
VII SEMESTER (Swappable VII and VIII SEMESTER) (SCHEME-A)															
Sl. No	Course and Course Code	Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)	Teaching Hours /Week				Examination				Credits			
				CI		LI	# TW & SL	# TH/S	D u r a t i	CIE Marks	SEE Marks		Total Marks		
				L	T									P	

										o n l i n e h o u r s				
1	IPCC	1BML701	Edge computing with Tiny ML	Respective Engg Dept	42	0	28	50	120	3	50	50	100	4
2	PEC	1BML702x	Professional Elective Course-III	Respective Engg Dept	42	0	0	48	90	3	50	50	100	3
3	PEC	1BML703x	Professional Elective Course -IV	Respective Engg Dept	42	0	0	48	90	3	50	50	100	3
4	OEC	1BML704x	Open Elective Course-I	Respective Engg Dept	42	0	0	48	90	3	50	50	100	3
5	SDC	1BML705	Capstone Project - Phase-II	Respective Engg Dept	0	0	0	210	210	3	100	100	200	7
6	NCMC	1BIKS706	Indian Knowledge System	TD/PSB	14	0	0	14	28	---	100	--	100	PP
	Total								628	15	400	300	700	20
	Professional Elective Course-III													
1BML702A		Medical Robotics		1BML702C		Speech Processing								
1BML702B		Physiological System Modelling		1BML702D		Biomechanics								
	Professional Elective Course-IV													
1BML703A		Wearable Technology		1BML703C		Brain Computer Interface								
1BML703B		BIO-MEMS		1BML703D		Medical Device Regulations								
	Open Elective Course-I													
1BML704A		IoT in Healthcare		1BML704C		Hospital Management								
1BML704B		Mechatronics in Medicine		1BML704D		Foreign Language (NPTEL/SWAYAM/online VTU)								

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Scheme of Teaching and Examinations 2025														
Outcome-Based Education (OBE) and Choice-Based Credit System (CBCS) (Effective from the academic year 2025-26)														
VIII SEMESTER (Swappable VII and VIII SEMESTER) (SCHEME-A)														
Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)	Teaching Hours /Week				Examination					
					CI		LI	# T W & S L	# TH/S	Duration in hours	CIE Marks	SEE Marks	Total Marks	Credits
1	PEC	1BML801x	Professional Elective-V (NPTEL/VTU Online Course)	Online Evaluation	42	0	0	48	90	3	50	50	100	3
2	OEC	1BML802x	Open Elective-II (NPTEL/VTU Online Course)	Online Evaluation	42	0	0	48	90	3	50	50	100	3
3	SDC	1BML803x	Internship (15 weeks or 90 working days)	--	--	--	--	360	3	100	100	200	09	
Total									540	9	200	200	400	15
Professional Elective Course (Online courses)-V														
1BML801A		NPTEL/VTU Online Course		1BML801C		NPTEL/VTU Online Course								
1BML801B		NPTEL/VTU Online Course		1BML801D		NPTEL/VTU Online Course								
Open Elective Courses -II (Online Courses)														
1BML802A		NPTEL/VTU Online Course		1BML802C		NPTEL/VTU Online Course								
1BML802B		NPTEL/VTU Online Course		1BML802D		Foreign Language (NPTEL/SWAYAM/online VTU)								
Types of Internships (Course Code: 1BML803x)														
Students shall undertake one of the following internship types during the eighth semester, as per academic guidelines:														
1. 1BML803A – Industry Internship: Shall involve practical exposure and training within an industrial or corporate setting.														
2. 1BML803B – Research Internship: Shall focus on academic or applied research under the guidance of faculty or research institutions.														
3. 1BML803C – Post-Placement Internship: Shall be undertaken by students who have secured placement, aligning with their future employment domain.														
4. 1BML803D – Societal Internship: Shall engage students in community-based or social impact projects with NGOs, government bodies, or civic organizations.														
5. 1BML803E – Online Internship: Shall be conducted through recognized digital platforms offering structured internship modules.														
6. 1BML803F – Skill Enhancement Internship: Shall be opted by students unable to secure internships, offering credit equivalence through curated online courses available at http:// www.online.vtu.ac.in														

	To ensure uniformity, quality, and transparency in the internship process, VTU has launched a centralized web portal that serves as a single platform for all internship opportunities. Reputed industries, Centres of Excellence, Research Laboratories , and other recognized bodies will be registered on this portal. Students must choose internships exclusively through this portal. No other mode of internship selection will be permitted
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VII and VIII semesters for the candidates who opt for a two-semester internship along with Capstone Project (Scheme B)														
Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)	Teaching & Learning Scheme				#-TH / S	Examination				Credits
					CI		LI	# TW & SL		Dur a t i o n i n h o u r s	CIE Marks	SEE Marks	Total Marks	
					L	T	P							
1	IPCC	1BML701	To be completed in the Summer Semester after 6 th semester						3	50	50	100	4	
2	PEC	1BML702x	Professional Elective Course-III (NPTEL/VTU Online Course)	Online Evaluation					--	50	50	100	3	
3	PEC	1BML703x	Professional Elective Course-IV (NPTEL/VTU Online Course)	Online Evaluation					--	50	50	100	3	
4	OEC	1BML704x	Open Elective Course (NPTEL/VTU Online course)-I	Online Evaluation					---	50	50	100	3	
5	SDC	1BML705	Capstone Project - Phase-II***	TD: PSB:					3	100	100	200	7	
6	NCMC	1BIKS706	Indian Knowledge System (VTU online)		1	0	0	0	--	100	--	100	PP	
	Total									6	400	300	700	20
1	PEC	1BML801x	Professional Elective-V (NPTEL/VTU Online Course)	Online Evaluation					---	50	50	100	3	
2	OEC	1BML802x	Open Elective-II (NPTEL/VTU Online Course)	Online Evaluation					---	50	50	100	3	
3	SDC	1BML803x	Internship (Two- semester internship for a minimum Period of 180 working days or -30 weeks)						3	50	50	100	09	
	Total									3	150	150	300	15

# TH/S- Total Hours per Semester, TW & SL- Term Work & Self Learning				
7 th semester and 8 th semester Credits Total				38
NPTEL/VTU Online Professional Elective Course - III				
1BML703A	NPTEL/VTU Online Courses	1BML703C		NPTEL/VTU Online Courses
1BML703B	NPTEL/VTU Online Courses	1BML704D		NPTEL/VTU Online Courses
NPTEL/VTU Online Open Elective Courses - I				
1BML704A	NPTEL/VTU Online Courses	1BML704C		NPTEL/VTU Online Courses
1BML704B	NPTEL/VTU Online Courses	1BML704D		NPTEL/VTU Online Courses
NPTEL/VTU Online Professional Elective Course (Online Courses)-IV				
1BML801A	NPTEL/VTU Online Courses	1BML801C		NPTEL/VTU Online Courses
1BML801B	NPTEL/VTU Online Courses	1BML801D		NPTEL/VTU Online Courses
NPTEL/VTU Online Open Elective Courses (Online Courses)-III				
1BML802A	NPTEL/VTU Online Courses	1BML802C		NPTEL/VTU Online Courses
1BML802B	NPTEL/VTU Online Courses	1BML802D		Foreign Language (NPTEL/MOOC/online VTU)
Types of Internships (Course Code: 1BML803) Students shall undertake one of the following internship types during the eighth semester, as per academic guidelines: 1BML803A – Industry Internship: Shall involve practical exposure and training within an industrial or corporate setting. 1BML803B – Research Internship: Shall focus on academic or applied research under the guidance of faculty or research institutions. 1BML803C – Post-Placement Internship: Shall be undertaken by students who have secured placement, aligning with their future employment domain. 1BML803D – Societal Internship: Shall engage students in community-based or social impact projects with NGOs, government bodies, or civic organizations. 1BML803E – Online Internship: Shall be conducted through recognized digital platforms offering structured internship modules. 1BML803F – Skill Enhancement Courses (SEC): Shall be opted by students unable to secure internships, offering credit equivalence through curated online courses available at http:// www.online.vtu.ac.in To ensure uniformity, quality, and transparency in the internship process, VTU has launched a centralized web portal that serves as a single platform for all internship opportunities. Reputed industries, Centres of Excellence, Research Laboratories , and other recognized bodies will be registered on this portal. Students must choose internships exclusively through this portal. No other mode of internship selection will be permitted.				



Overview of Courses, Credits, Projects, and Internships under VTU Curriculum

I. Abbreviations used in the Scheme of Teaching and Examinations

Abbreviations	Expanded Form of the Abbreviations
AICTE	All India Council of Technical Education
NCrF	National Credit Framework
VTU	Visvesvaraya Technological University
AEC	Ability Enhancement Course
ASC	Applied Science Course
BSC	Basic Science Course
CIE	Continuous Internal Evaluation
CI	Classroom Instruction
CCA	Continuous Comprehensive Assessment
CGPA	Cumulative Grade Point Average
CUL	Cultural
COE	Centre for Online Education
HSMC	Humanities Studies and Management Course
IPCC	Integrated Professional Core Course
LI	Laboratory Instruction
L	Lecture
NCMC	Non-Credit Mandatory Course
NSS	National Service Scheme
NPTEL	National Programme for Technical Enhanced Learning
OEC	Open Elective (Interdepartmental or interdisciplinary) Course
PCC	Professional Core Course
PCCL	Professional Core Course Laboratory
PEC	Professional Elective Courses

PE	Physical Education
P	Practical
SEC	Skill Enhancement Courses
SEE	Semester End Evaluation
SL	Self-Learning
SGPA	Semester Grade Point Average
SWAYAM	Study Webs of Active-Learning for Young Aspiring Minds
TW	Term Work
T	Tutorial
VTU online Course	VTU online courses offered by Centre for Online Education, Mysuru
YOG	Yoga

II. Credit Representation

1-hour Lecture (L) per week=1Credit

2-hours Tutorial(T) per week=1Credit

2-hours Practical / Drawing (P) per week=1Credit

Teaching & Learning Scheme

As per the new National Credit Framework (NCrF), 30 hours of learning of a student is considered equivalent to 1 credit. A semester is considered as a 14-week period of academic interaction with students. The learning components are categorized as follows:

- Classroom Instruction (CI):** Includes different instructional / implementation strategies i.e. Lecture (L), Tutorial (T), Case method, Demonstrations, Video demonstration, Problem based learning etc. to deliver theoretical concepts within the classroom measured in Number of hours per semester.
- Laboratory Instruction (LI):** Expressed as number of hours per semester which Includes experiments / practical performances / problem-based experiences in laboratory, workshop, field or other locations using different instructional / Implementation strategies.
- Term work (TW):** Includes assignments, seminars, presentations, case studies, micro projects, field activities, industrial visits, academic preparation duration and any other student activities in Number of hours per semester.
- Self-Learning (SL):** MOOCs (SWAYAM/NPTEL/Industry certified courses), spoken tutorials, online educational resources, self-initiated projects, Learning through digital resources etc in Number of hours per semester. (If provided in curriculum structure).

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	Course Details	
1.	One Credit Theory Courses:	
	Teaching-Learning sessions in a semester	14 hours
	Examination pattern for CIE and SEE	Multiple Choice Question (MCQ)
	Teaching hours per week - L:T:P	1:0:0
2.	One Credit Laboratory Courses:	
	Teaching-Learning sessions in a semester	28 hours (2 hours session /week)
	Examination pattern for CIE and SEE	Continuous assessments, lab Internal test and SEE
	Teaching hours per week - L:T:P	0:0:2
3.	Two Credit Theory Courses:	
	Teaching-Learning Sessions in a semester	28 hours
	Examination pattern for CIE and SEE	Descriptive
	Teaching hours per week - L:T:P	2:0:0
4.	Three Credit ESC/ETC/PCC/PEC/OEC Courses:	
	Teaching-Learning Sessions in a semester	42 hours
	Examination pattern for CIE and SEE	Descriptive
	Teaching hours per week for theory courses - L:T:P	3:0:0
5.	Four Credit Program Core Courses (PCC):	
	Teaching-Learning Sessions in a semester	56 hours
	Examination pattern for CIE and SEE	Descriptive
	Teaching hours per week for theory courses - L:T:P	4:0:0
6.	Four Credit Integrated Professional Core Courses (IPCC):	
	Teaching-Learning Sessions in a semester (Teaching sessions: 42 hours + Practical sessions: 28 hours)	70 hours
	Examination pattern for CIE and SEE	Descriptive

Practical part of examination	CIE (No SEE).
Teaching hours per week - L: T: P	3: 0: 2

III. Details of Courses

- (1) Integrated Professional Core Course (IPCC):** The Integrated Professional Core Course (IPCC) refers to a core theory course that is integrated with a laboratory of the same subject. Each IPCC carries 4 credits, with Teaching–Learning hours structured (L : T : P) as either (3:0:2). The theory component of the IPCC shall be evaluated through both Continuous Internal Evaluation (CIE) and Semester End Examination (SEE). The laboratory part shall be assessed exclusively through CIE, with no SEE. However, questions derived from the laboratory part may be included in the SEE question paper to ensure comprehensive evaluation
- (2) Non-Credit Mandatory Courses (NCMC):** are aimed at enhancing students' knowledge, skills, and awareness beyond the core curriculum. Successful completion of the NCMC is compulsory for fulfilling the requirements of the academic program. It shall not be considered for the computation of SGPA, CGPA and vertical progression. Each student shall register for the prescribed NCMC(s) in the prescribed semester. A student who fails to qualify in the prescribed NCMC shall not be eligible for the conferment of the degree.
- (3) Professional Elective Courses (PEC):** A professional elective course (PEC) is intended to enhance the depth and breadth of educational experience in the Engineering and Technology curriculum of the same discipline.
- (4) Open Elective Courses (OEC):** A open elective course (OEC) is a course offered by departments other than a student's parent department. These interdepartmental /interdisciplinary courses allow students to explore disciplines beyond their core area of study. These courses are intended to promote interdisciplinary learning, broad-based education, thereby enhancing a student's overall knowledge, creativity, and employability. Registration to open electives shall be documented under the guidance of the Program Coordinator/ Advisor/Mentor/Proctor.
- (5) Ability Enhancement Course Laboratory (AEC):** An Ability Enhancement Course Laboratory is a practical, skill-oriented lab course designed to strengthen students' practical abilities, professional competencies that support communication, environmental awareness, computational thinking, interdisciplinary learning, and application skills through hands-on learning experiences.

The laboratory may pertain to disciplinary or interdisciplinary involving experiments, design tasks, and mini-projects aligned with current industry practices.

- (6) Skill Enhancement Courses (SEC):** These courses are intended to develop specific practical skills and competencies that improve students' employability, technical proficiency, and professional readiness to bridge the gap between academic and industry requirements. These courses emphasize hands-on training, application of theoretical knowledge, and development of discipline-relevant and transferable skills required in industry and society, and develop entrepreneurship and start-up skills.
- (7) Online Courses:** Online courses are educational programs delivered over the Internet through a digital platform, allowing students to access lessons, assignments, and discussions from anywhere at any time. Most online courses offer flexibility, allowing students to access materials and complete assignments on their own schedule. However, students have to pass the course within a stipulated period as per the norms of the university.
- (8) VTU Online Courses:** VTU Online courses are online courses offered by Centre of Online Education (COE) Mysuru. A wide range of multidisciplinary courses are available to learners anywhere, anytime to earn university-prescribed credits through proctored examination for the award of a degree.
- (9) NPTEL/SWAYAM Online Courses:** The National Programme on Technology Enhanced Learning (NPTEL)/SWAYAM (Study Webs of Active Learning for Young Aspiring Minds) are the specific Indian platforms to host national Massive Open Online Courses (MOOCs). It offers online courses on a wide range of disciplines to learners anywhere, anytime, to earn university-prescribed credits through proctored examination for the award of a degree. All NPTEL/SWAYAM courses are MOOCs, but not all MOOCs are offered on these specific Indian platforms.

IV. **National Service Scheme / Physical Education / Yoga (NSS / PE / YOG):**

All students are required to register for any one of the following courses; National Service Scheme (NSS), Physical Education (PE) (Sports and Athletics), or Yoga (YOG)—with the respective course coordinator during the first week of the third semester.

- Colleges shall submit Continuous Internal Evaluation (CIE) marks for each semester based on the activities completed by students under the selected course.
- Students may opt for different activities/options across semesters. For instance, a student participating in PE during 3rd semester may choose NSS in the 4th semester or Yoga.
- Activities shall be conducted over two semesters (III & IV), and successful completion of the registered course / or courses along with the required CIE score is mandatory for the award of the degree.
- Institutions must ensure that events are appropriately scheduled and reflected in the semester-wise calendar for NSS, PE, Music, and Yoga activities.

These courses shall not be considered for the calculation of SGPA or CGPA and for vertical progression. However, completion of course(s) is compulsory for degree eligibility.

V. Projects

1. Community Project

A community is a social unit or group of people sharing socially-significant characteristics, such as place, set of norms, culture, religion, values, customs or identity. A community project involves addressing issues or needs within such a community or a network of entities working toward a common purpose. These projects may cover a wide range of areas, including welfare, sustainability, technology integration, and social development. Examples include establishing and maintaining an orphanage, implementing solar power generation and its maintenance, or developing environmental improvement solutions, etc. A community project is an experiential learning activity that encourages students to identify, analyse, and address real-life problems of the community using engineering knowledge. It aims to promote social responsibility and civic engagement, interdisciplinary thinking and collaboration and practical application of theoretical concepts, thereby enabling students to contribute meaningfully to community welfare and sustainable development. Students can take up project individually or in a group not exceeding 4 students.

The evaluation shall be done as per the following;

CIE: The CIE marks shall be awarded by a committee consisting of the Head of the concerned Department and two senior faculty members of the Department, one of whom shall be the Guide. The CIE marks awarded for the project work shall be based on the rubrics.

SEE: SEE will be conducted by the two examiners appointed by the University. The SEE marks awarded for the project work shall be based on the rubrics.

2. Environmental Science Project

The Environmental Science Project is an applied learning component designed to develop students' awareness, understanding, and responsibility toward the environment. It provides an opportunity to study real-world environmental issues and apply scientific and engineering principles to design feasible and sustainable solutions.

The topics under environment include, but not limited to, climate change, biodiversity, air and water pollution, land use, excess use of natural resources, earthquakes, rise in the earth's temperature, power generation, soil erosion, environment issues related programme, etc.

The project involves problem identification, field surveys, case studies, data collection, environmental audits, analysis, and proposal of remedial or preventive measures aimed at improving biodiversity, air quality, and thermal comfort, etc. Students can take up project individually or in a group not exceeding 4 students. Students can opt for Interdisciplinary Project based on their interest.

The evaluation shall be done as per the following;

CIE: The CIE marks shall be awarded by a committee consisting of the Head of the concerned Department and two senior faculty members of the Department, one of whom shall be the Guide. The CIE marks awarded for the project work, shall be based on the rubrics.

SEE: SEE will be conducted by the two examiners appointed by the University. The SEE marks awarded for the project work shall be based on the rubrics.

3. Hackathon Based Project (Academic)

The term hackathon is derived from the combination of hack (referring to clever problem-solving, not illegal activity) and marathon, which denotes an arduous (i.e., difficult) intellectual task requiring sustained effort, endurance, and mental resilience. The meaning of a hackathon varies depending on the specific context and intent. In an academic context, a hackathon can be considered to involve several concepts, ranging from resourceful, unconventional approaches to problem-solving.

Though a hackathon is an event, typically lasting for a few days to address a specific challenge, for academic purposes, it is conducted as a noncompetitive semester-long activity. The evaluation is done as and when the project is completed, by a panel of industry experts.

The hackathons not only help participants develop skills like problem-solving, critical thinking, creativity, teamwork, communication and time management, but also foster indigenous technology development, promote innovation and entrepreneurship, and contribute to non-formal learning and skill enhancement.

Students can take up a hackathon project individually or in a group of not exceeding 4 students.

The respective **BoS will announce** the problem statements in the beginning of the 5th semester. The topic selected can be discipline specific, interdepartmental, industrial, social (refers to immediate human relations, interactions, and individual behaviour within a community), societal (describes larger, general issues, institutions, and structures that define society as a whole), environmental, health, financial, or innovative in nature, leading to development of a working prototype, application, or product.

Hackathon projects are aligned with the principles of Outcome-Based Education (OBE) and support the objectives of innovation, skill development, and experiential learning in engineering education.

Projects shall be evaluated by industry experts, based on creativity, problem-solving approach, teamwork, and possible implementation, as far as possible, as and when the project is completed.

The evaluation shall be done as per the following;

CIE: The CIE marks shall be awarded by a committee consisting of the Head of the concerned Department and two senior faculty members of the Department, one of whom shall be the Guide. The CIE marks awarded for the project work, shall be based on the rubrics.

SEE: SEE will be conducted by the industry experts appointed by the Head of the Institute/University. The SEE marks awarded for the project work shall be based on the rubrics.

4. Capstone Project

The Capstone project is a comprehensive, year-long project carried out in two phases during 6th and 7th semesters of the undergraduate engineering/technology program. It integrates knowledge and skills acquired from multiple courses and disciplines to address a complex, real-world problem.

This project provides students with an opportunity to apply scientific principles, engineering methodologies, and technological tools to conceive, design, implement and evaluate an engineering solution. It serves as a culminating academic experience to demonstrate program outcomes, including problem-solving ability, teamwork, communication skills, and practical application of engineering principles. Students can take up project individually or in a group not exceeding 4 students. The group may have students from the same discipline and drawn from different disciplines.

Types of Capstone Projects:

Capstone projects undertaken for one year may fall into one or more of the following categories:

a) **Research-Oriented Projects :**

- Focus on investigating new concepts, theories, or technologies.
- Aim to generate new knowledge or contribute to academic research.

b) **Experimental/Analytical Projects**

- Based on laboratory or field experiments to validate a hypothesis or study a phenomenon.
- Including detailed data collection, analysis, and interpretation.

c) **Simulation/Modelling Projects**

- Use computational tools to model, simulate, and predict system behaviour.
- Reduce the need for physical prototyping in the initial stages.

d) **Industrial/Industry-Sponsored Projects**

- Carried out in collaboration with an industry partner.
- Address real-world engineering problems faced by the organization.

e) **Interdisciplinary/Multidisciplinary Projects**

- Combine knowledge and techniques from multiple engineering domains or other fields such as management, medicine, or environmental sciences.

f) **Entrepreneurial/Innovation Projects**

- Focus on product or service innovation with potential for commercialization.

- Include aspects of market analysis, cost estimation, and business planning.

Phase I Evaluation: Capstone Project Phase-I shall have only Continuous Internal Evaluation (CIE). In case disciplinary capstone project, the CIE shall be conducted by the **Departmental Project Review Committee**, which consists of a Senior Professor, the Project Guide, and one additional faculty member appointed by the principal for projects within the **parent discipline**.

For **Interdisciplinary Projects**, the Project Review Committee will consist of one Senior Professor, the department and interdepartmental Project Guides and one faculty member from a department related to the interdisciplinary project. The committee members are appointed by the principal of the college.

Phase-I evaluation shall be based on **rubrics** designed to measure graduate attributes defined by NBA. Successful completion of Phase-I allows the student to proceed to **Phase-II**.

Phase II Evaluation:

CIE of Phase shall be evaluated as indicated with phase -I evaluation. The **SEE** shall be conducted by university-appointed examiners. The assessment shall be based on **rubrics** designed to measure graduate attributes defined by NBA.

VI. Internship

Internship refers to the position of a student as trainee or a temporary (or unconfirmed) employee, who works in an organization, with or without pay, in order to gain work experience or satisfy requirements for a qualification. It is a structured, supervised professional experience in an industry, research organization, or community setting. Students taking up internship may be with or without stipend. Internships play a vital role in bridging the gap between theoretical education and professional practice. In general, engineering internships serve as a crucial component of professional education by providing experiential learning, industry readiness, and holistic skill development, ultimately producing competent engineers or entrepreneurs. Apart from these, it develops professional ethics, work culture awareness and communication skills.

Some of the common types of internships are as follows:

- i. **Industry Internship:** Carried out in the engineering industry, companies, manufacturing units, startups, business, IT industry. The topic involved may be technical, managerial, production-related tasks, live projects, or innovative activities.
- ii. **Research Internship:** Carried out at universities, research labs, or R and D departments or organisations. The internship may involve literature review, data analysis, and experimental work leading to publications, prototypes, technical reports or innovations. The research internship may induce students to plan for higher studies or academic careers.
- iii. **Academic or Teaching Internship:** Carried out at educational institutions. The students assist in academic activities, laboratory sessions or content development, and prepare or present report, presentation and student evaluation. The internship encourages interest in academia and pedagogy, develops new skills, helps to gain a competitive edge on the job market or for post-baccalaureate studies.
- iv. **Community or Societal Internship:** Carried out with government schemes, or rural development projects, Non-Governmental Organisations (NGOs). The internship focused on social and community development activities promotes social responsibility, sustainable development awareness, encourages civic responsibility and ethical engagement.
- v. **Entrepreneurship Internship:** Undertaken in association with start-ups, or entrepreneurship cells or launching own idea in Pre-Incubation/Incubation centres. The internship offers exposure to business planning, prototype product development, and promotes innovation, risk-taking, and entrepreneurial mindset.
- vi. **Virtual or Remote or Online Internship:** Undertaken using online tools and digital collaboration platforms. Such internships are common in content writing, data science, marketing, and software development. It offers flexible learning environments and access to global opportunities, and allows participation in real projects without being physically present, from anywhere and anytime.
- vii. **Government Internship:** Ministries, public sector units, or civic bodies offer such internships in policy research, administrative tasks, or public service projects. This internship is for students interested in governance or public administration.
- viii. **Post-Placement Internship:** Refers to the internship offered to students after they receive a confirmed job offer (placement) from a company, but before formally joining as full-time employees. This internship (on-site, virtual, or hybrid) ensures that students are

groomed to be professionally ready, technically competent, and culturally aligned with the organization even before official induction.

- ix. **Skill Enhancement Internship:** Carried out at reputed organisations in offline or online mode. The aim of the internship is to expose to real-world tools, technologies, and professional environments to improve a student's employability by offering hands-on experience, application of theoretical concepts, and skill development aligned with current industry and technical trends. Skill Enhancement Internships, depending on focus area and scope, can be carried out at various organisations such as, Academic and Research Institutions, Industry and Corporate Settings, Government and Public Sector, NGOs and Social Enterprises.

For Skill Enhancement Internship topics refer to

<https://online.vtu.ac.in/category/courses/Skill-Enhancement-Course>.

Note on Internship for the Attention of Students and Colleges

- Placement training conducted at the college level, whether by third-party agencies, training institutes, or internal faculty, shall not be considered as internship for either a 15 week or a 30-week period.
- The official engagement period of 15-week or 30-week for students selected/recruited by the company/ organization only at their premises under the supervision of the company, shall only be considered as an internship.
- The period of training and working of students who have been recruited as employees by organisations at the beginning of the 4th year of the programme, shall also be treated as an internship.
- Students and colleges/institutions shall follow all the guidelines and procedures of the organization and the University's Internship Guidelines, and complete the internship within a period that matches with the VTU Calander and examination timetable.
- The assigned institution faculty mentor/ coordinator/guide should monitor the student's progress, and document offer letters, training reports, attendance, and evaluations for awarding academic credits.
- All students undergoing an internship, should adhere to all the guidelines, reporting protocols, and evaluation procedures prescribed by the University.

- Students must submit the certificate of completion of an internship with the period of internship clearly mentioned, from the respective company/organization.
- Colleges must submit details of students opting for internship during the odd and even semesters, along with a copy of the company selection letter, to the VTU when notified by the University.

Attention: In addition to the internship support provided by the college, students have the option to select internships through the AICTE and VTU Internship Portals. To ensure uniformity, quality, and transparency in the internship process, VTU has developed a dedicated web portal that serves as a single platform where colleges can also register companies offering internships. Every student is required to register on the portal before the commencement of their internship, and their progress will be monitored through the same platform.

As per VTU norms, the CIE shall be conducted based on the students' performance during the training program, assessed through **rubrics** from the company supervisor. The SEE evaluation shall be conducted by the college as per the examination timetable published by the VTU.

VII. Bridge Courses on Mathematics for Lateral Entry Students:

This courses can be taught in the **offline** mode by the faculty of the mathematics department of the college as per the normal procedure to the students. The students can attend the class at their college or the can choose the VTU **online mode**, conducted by Centre for Online Education (COE) of VTU. Only CIE is only prescribed for this course and the CIE assessment is only by VTU online COE, and not at the institution level.

All lateral entry students are required to **register** compulsorily for this course in the 3rd semester & 4th Semester and must appear for **CIE**. Passing in this course is **mandatory** for the award of the degree. Those who fail to secure the passing CIE marks, have to appear for

the summer semester of the academic year or during subsequent odd semester. However, this course will not be considered for vertical progression, SGPA, and CGPA calculation.

VIII. AICTE Activity Points Requirement for BE/B.Tech. Programmes

As per AICTE guidelines (refer to Chapter 6 – *AICTE Activity Point Program, Model Internship Guidelines*), in addition to academic requirements, students must earn a specified number of **Activity Points** to be eligible for the award of the degree. The points to be earned are as follows:

1. **Regular students** admitted to a 4-year degree program must earn **100 Activity Points**.
2. **Lateral entry students** (joining from the second year) must earn **75 Activity Points**.
3. **Students transferred** from other universities directly into the fifth semester must earn **50 Activity Points** from the date of entry into VTU.

These Activity Points do not carry any credits, and therefore, the points are not considered for **the SGPA/CGPA** or for **vertical progression**. However, earning Activity Points is mandatory for the **award of the degree**, and the points earned will be reflected on the **eighth semester Grade Card**.

The hours spent earning the activity points will not be counted for regular attendance requirements. Students can accumulate these points at any time during their program period, including weekends, holidays, and vacations, starting from the year of admission, provided they meet the minimum hours of engagement prescribed for each activity by AICTE.

If a student completes all the semesters (eight/six) successfully, but fails to earn the required Activity Points, the eighth-semester Grade Card will be withheld until the Activity points requirement is fulfilled. Also, the degree will be awarded only after the Grade Card has been released.

IX. Option -1: Swappable Semester Scheme - A

To ensure equitable access to internship opportunities, provision has been made to swap seventh and eighth semesters under Scheme

A. The details of the Scheme – A are as follows:

- Students who have an offer to enrol for a 15-week internship, before the start of 4th year, shall register for VIII semester courses instead of VII semester courses and take up respective semester examination.
- Those who have no offer to enrol to a 15-week internship, before the start of 4th year, shall register VII and VIII semesters courses in the chronological manner and complete the programme. In this case the internship shall be carried out during VIII semester.

X. Option -2: Two-Semester Internship Scheme – B

- Students who have cleared all the courses up to VI semester in first attempt only (i.e., students having no backlogs) and have an internship offer for a period of 180 working days or 30 working weeks, are only eligible for Scheme – B. The internship commence date should coincide with the 4th year academic calendar of VTU. Such students, shall produce the confirmed internship letter, to the Principal/Academic Authority to get permission to register for the summer semester to opt for Scheme - B.
- Such eligibles students shall register for the course 1BML701 in the summer semester of the same academic year (i.e., after their VI semester) and complete the said course in first attempt only.
- In case, they absent for the examination or fails in the course 1BML701, they shall not be considered eligible for the Scheme – B. However, they shall register for Scheme – A.
- After completing the course 1BML701, students with confirmed internship letter to carry out the internship for a minimum 180 working days or 30 working weeks, shall register for the Scheme – B.
- In case students cannot commence the internship for various reasons, they not be considered for Scheme – B. In such cases, they shall register for Scheme – A. However, they will be exempted from studying the course 1BML701 again.
- A request letter with internship permission letter must be submitted to Registrar, VTU through concerned authorities of the institution. Only after receiving the approval from the Registrar, students proceed with the internship as mentioned in Option Scheme B.

Capstone Project Evaluation Guidelines for Students Opting for Internship for two semesters duration:

- a) **Industry Internship Leading to Capstone Project:** For students opting for a two-semester Industry Internship that leads to the completion of the Capstone Project, the Phase-I evaluation will be conducted at the end of the VII semester, and the Phase-II evaluation will be conducted at the end of the VIII semester.
- b) **Industry Internships Not Leading to Capstone Project:** For students opting for a Industry Internship that does **not** lead to the completion of the Capstone Project, they are required to undertake the Capstone Project separately. Both Phase-I and Phase-II of the Project Work must be completed as per the prescribed guidelines, under the guidance of a college-level guide or mentor.

